

REPORT ON WIRESHARK CAPTURE FROM A PCAP FILE

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Executive Summary

This network traffic capture documents a high-volume **TCP SYN Flood Attack** targeting an internal web server. The attack successfully overwhelmed the host, causing legitimate user connections to time out. The log is colour-coded and enclosed in the folder containing this report. The colour key is as follows:

a) green: Normal TCP handshakes b) red: Attack activity c) yellow: normal connections interrupted by attacks

Host Profiles

- **Target Server:** 192.0.2.1 (Listening on port 443 for HTTPS traffic).
- **Primary Threat Actor:** 203.0.113.0 .
- **Legitimate Clients:** Multiple IPs including 198.51.100.23, 198.51.100.14, 198.51.100.5, 198.51.100.16, and 198.51.100.7.

See the images below depicting the information provided above:

Color Coded TCP Log						
Color Key:						
green	Normal TCP connection handshakes					
red	Attack activity					
yellow	Normal TCP connections failing due to attack					
Color as text	No.	Time (in seconds & milliseconds)	Source	Destination	Protocol	Info
green	47	3.144521	198.51.100.23	192.0.2.1	TCP	42584->443 [SYN] Seq=0 Win=5792 Len=120...
green	48	3.195755	192.0.2.1	198.51.100.23	TCP	443->42584 [SYN, ACK] Seq=0 Win=5792 Len=120...
green	49	3.246989	198.51.100.23	192.0.2.1	TCP	42584->443 [ACK] Seq=1 Win=5792 Len=120...
green	50	3.298223	198.51.100.23	192.0.2.1	HTTP	GET /sales.html HTTP/1.1
green	51	3.349457	192.0.2.1	198.51.100.23	HTTP	HTTP/1.1 200 OK (text/html)
red	52	3.390692	203.0.113.0	192.0.2.1	TCP	54770->443 [SYN] Seq=0 Win=5792 Len=0...
red	53	3.441926	192.0.2.1	203.0.113.0	TCP	443->54770 [SYN, ACK] Seq=0 Win=5792 Len=120...
red	54	3.49316	203.0.113.0	192.0.2.1	TCP	54770->443 [ACK Seq=1 Win=5792 Len=0...
green	55	3.544394	198.51.100.14	192.0.2.1	TCP	14785->443 [SYN] Seq=0 Win=5792 Len=120...
green	56	3.599628	192.0.2.1	198.51.100.14	TCP	443->14785 [SYN, ACK] Seq=0 Win=5792 Len=120...
red	57	3.664863	203.0.113.0	192.0.2.1	TCP	54770->443 [SYN] Seq=0 Win=5792 Len=0...
green	58	3.730097	198.51.100.14	192.0.2.1	TCP	14785->443 [ACK] Seq=1 Win=5792 Len=120...
red	59	3.795332	203.0.113.0	192.0.2.1	TCP	54770->443 [SYN] Seq=0 Win=5792 Len=120...
green	60	3.860567	198.51.100.14	192.0.2.1	HTTP	GET /sales.html HTTP/1.1
red	61	3.939499	203.0.113.0	192.0.2.1	TCP	54770->443 [SYN] Seq=0 Win=5792 Len=120...
green	62	4.018431	192.0.2.1	198.51.100.14	HTTP	HTTP/1.1 200 OK (text/html)
green	63	4.097363	198.51.100.5	192.0.2.1	TCP	33638->443 [SYN] Seq=0 Win=5792 Len=120...
red	64	4.176295	192.0.2.1	203.0.113.0	TCP	443->54770 [SYN, ACK] Seq=0 Win=5792 Len=120...
green	65	4.255227	192.0.2.1	198.51.100.5	TCP	443->33638 [SYN, ACK] Seq=0 Win=5792 Len=120...
red	66	4.256159	203.0.113.0	192.0.2.1	TCP	54770->443 [SYN] Seq=0 Win=5792 Len=0...
green	67	5.235091	198.51.100.5	192.0.2.1	TCP	33638->443 [ACK] Seq=1 Win=5792 Len=120...
red	68	5.236023	203.0.113.0	192.0.2.1	TCP	54770->443 [SYN] Seq=0 Win=5792 Len=0...
green	69	5.236955	198.51.100.16	192.0.2.1	TCP	32641->443 [SYN] Seq=0 Win=5792 Len=120...
red	70	5.237887	203.0.113.0	192.0.2.1	TCP	54770->443 [SYN] Seq=0 Win=5792 Len=0...
green	71	6.228728	198.51.100.5	192.0.2.1	HTTP	GET /sales.html HTTP/1.1

red	72	6.229638	203.0.113.0	192.0.2.1	TCP	54770->443 [SYN] Seq=0 Win=5792 Len=0...
yellow	73	6.230548	192.0.2.1	198.51.100.16	TCP	443->32641 [RST, ACK] Seq=0 Win=5792 Len=120...
red	74	6.330539	203.0.113.0	192.0.2.1	TCP	54770->443 [SYN] Seq=0 Win=5792 Len=0...
green	75	6.330885	198.51.100.7	192.0.2.1	TCP	42584->443 [SYN] Seq=0 Win=5792 Len=0...
red	76	6.331231	203.0.113.0	192.0.2.1	TCP	54770->443 [SYN] Seq=0 Win=5792 Len=0...
yellow	77	7.330577	192.0.2.1	198.51.100.5	TCP	HTTP/1.1 504 Gateway Time-out (text/html)
red	78	7.351323	203.0.113.0	192.0.2.1	TCP	54770->443 [SYN] Seq=0 Win=5792 Len=0...
green	79	7.360768	198.51.100.22	192.0.2.1	TCP	6345->443 [SYN] Seq=0 Win=5792 Len=0...
yellow	80	7.380773	192.0.2.1	198.51.100.7	TCP	443->42584 [RST, ACK] Seq=1 Win=5792 Len=120...
red	81	7.380878	203.0.113.0	192.0.2.1	TCP	54770->443 [SYN] Seq=0 Win=5792 Len=0...
red	82	7.383879	203.0.113.0	192.0.2.1	TCP	54770->443 [SYN] Seq=0 Win=5792 Len=0...
red	83	7.482754	192.0.2.1	203.0.113.0	TCP	443->54770 [RST, ACK] Seq=1 Win=5792 Len=0...
red	84	7.581629	203.0.113.0	192.0.2.1	TCP	54770->443 [SYN] Seq=0 Win=5792 Len=0...
yellow	85	7.680504	192.0.2.1	198.51.100.22	TCP	443->6345 [RST, ACK] Seq=1 Win=5792 Len=0...
red	86	7.709377	203.0.113.0	192.0.2.1	TCP	54770->443 [SYN] Seq=0 Win=5792 Len=0...
red	87	7.738241	203.0.113.0	192.0.2.1	TCP	54770->443 [SYN] Seq=0 Win=5792 Len=0...
red	88	7.767105	203.0.113.0	192.0.2.1	TCP	54770->443 [SYN] Seq=0 Win=5792 Len=0...
red	89	13.895969	192.0.2.1	203.0.113.0	TCP	443->54770 [RST, ACK] Seq=1 Win=5792 Len=0...
red	90	13.919832	203.0.113.0	192.0.2.1	TCP	54770->443 [SYN] Seq=0 Win=5792 Len=0...
red	91	13.943695	203.0.113.0	192.0.2.1	TCP	54770->443 [SYN] Seq=0 Win=5792 Len=0...
yellow	92	13.967558	192.0.2.1	198.51.100.16	TCP	443->32641 [RST, ACK] Seq=1 Win=5792 Len=120...
red	93	13.991421	203.0.113.0	192.0.2.1	TCP	54770->443 [SYN] Seq=0 Win=5792 Len=0...
red	94	14.015245	203.0.113.0	192.0.2.1	TCP	54770->443 [SYN] Seq=0 Win=5792 Len=0...
red	95	14.439072	192.0.2.1	203.0.113.0	TCP	443->54770 [RST, ACK] Seq=1 Win=5792 Len=0...
red	96	14.862899	203.0.113.0	192.0.2.1	TCP	54770->443 [SYN] Seq=0 Win=5792 Len=0...
green	97	14.886727	198.51.100.9	192.0.2.1	TCP	4631->443 [SYN] Seq=0 Win=5792 Len=0...
red	98	15.310554	203.0.113.0	192.0.2.1	TCP	54770->443 [SYN] Seq=0 Win=5792 Len=0...
red	99	15.734381	203.0.113.0	192.0.2.1	TCP	54770->443 [SYN] Seq=0 Win=5792 Len=0...
red	100	16.158208	192.0.2.1	203.0.113.0	TCP	443->54770 [RST, ACK] Seq=1 Win=5792 Len=0...
red	101	16.582035	203.0.113.0	192.0.2.1	TCP	54770->443 [SYN] Seq=0 Win=5792 Len=0...
red	102	17.005862	203.0.113.0	192.0.2.1	TCP	54770->443 [SYN] Seq=0 Win=5792 Len=0...
red	103	17.429678	192.0.2.1	203.0.113.0	TCP	443->54770 [RST, ACK] Seq=1 Win=5792 Len=0...
red	104	17.452693	203.0.113.0	192.0.2.1	TCP	54770->443 [SYN] Seq=0 Win=5792 Len=0...
red	105	17.475708	203.0.113.0	192.0.2.1	TCP	54770->443 [SYN] Seq=0 Win=5792 Len=0...
red	106	17.498723	203.0.113.0	192.0.2.1	TCP	54770->443 [SYN] Seq=0 Win=5792 Len=0...

Incident Timeline & Analysis

1. Baseline Phase (Normal Operation)

- **Packets 47–51 (Time: 3.14s – 3.34s):** Client 198.51.100.23 establishes a normal TCP handshake and successfully fetches /sales.html via an HTTP 200 OK response.

2. Onset of Attack

- **Packet 52 (Time: ~3.39s):** The attacker (203.0.113.0) initiates a flood using a fixed source port (54770) targeting port 443.
- **Packets 55–62:** Legitimate traffic from clients 198.51.100.14 and 198.51.100.5 still successfully connects and downloads the web page, but the frequency of attacker SYN packets begins escalating rapidly.

3. Service Degradation & Denial of Service (DoS)

- **Packets 72–74 (Time: ~6.23s):** The server resource queue fills up. The server begins forcibly closing legitimate connection attempts from client 198.51.100.16 by sending [RST, ACK] packets
- **Packet 77 (Time: 7.33s):** The server fails to process client 198.51.100.5's request in time, resulting in an HTTP 504 Gateway Time-out error.
- **Packets 80–121:** Legitimate clients (198.51.100.7, 198.51.100.22, and 198.51.100.9) are systematically rejected with TCP resets (RST) as the server denies new incoming handshakes.

4. Sustained Flooding Phase

- **Packets 122–214 (Time: 20.16s – 51.82s):** The attacker maintains an uninterrupted barrage of [SYN] packets, effectively rendering the web service completely unavailable to legitimate users. The images below show the traffic overwhelm using [SYN] packets and the interrupted normal traffic in yellow.

red	107	17.521738	203.0.113.0	192.0.2.1	TCP	54770->443 [SYN] Seq=0 Win=5792 Len=0...
red	108	17.544753	203.0.113.0	192.0.2.1	TCP	54770->443 [SYN] Seq=0 Win=5792 Len=0...
red	109	17.567768	192.0.2.1	203.0.113.0	TCP	443->54770 [RST, ACK] Seq=1 Win=5792 Len=0...
red	110	17.590783	203.0.113.0	192.0.2.1	TCP	54770->443 [SYN] Seq=0 Win=5792 Len=0...
red	111	18.413795	203.0.113.0	192.0.2.1	TCP	54770->443 [SYN] Seq=0 Win=5792 Len=0...
red	112	18.436807	203.0.113.0	192.0.2.1	TCP	54770->443 [SYN] Seq=0 Win=5792 Len=0...
red	113	18.459819	203.0.113.0	192.0.2.1	TCP	54770->443 [SYN] Seq=0 Win=5792 Len=0...
red	114	18.482831	203.0.113.0	192.0.2.1	TCP	54770->443 [SYN] Seq=0 Win=5792 Len=0...
red	115	18.506655	203.0.113.0	192.0.2.1	TCP	54770->443 [SYN] Seq=0 Win=5792 Len=0...
red	116	18.529667	203.0.113.0	192.0.2.1	TCP	54770->443 [SYN] Seq=0 Win=5792 Len=0...
red	117	18.552679	192.0.2.1	203.0.113.0	TCP	443->54770 [RST, ACK] Seq=1 Win=5792 Len=0...
red	118	18.875692	203.0.113.0	192.0.2.1	TCP	54770->443 [SYN] Seq=0 Win=5792 Len=0...
red	119	19.198705	203.0.113.0	192.0.2.1	TCP	54770->443 [SYN] Seq=0 Win=5792 Len=0...
red	120	19.521718	203.0.113.0	192.0.2.1	TCP	54770->443 [SYN] Seq=0 Win=5792 Len=0...
yellow	121	19.844731	192.0.2.1	198.51.100.9	TCP	443->4631 [RST, ACK] Seq=1 Win=5792 Len=0...
red	122	20.167744	203.0.113.0	192.0.2.1	TCP	54770->443 [SYN] Seq=0 Win=5792 Len=0...
red	123	20.490757	203.0.113.0	192.0.2.1	TCP	54770->443 [SYN] Seq=0 Win=5792 Len=0...
red	124	20.81377	192.0.2.1	203.0.113.0	TCP	443->54770 [RST, ACK] Seq=1 Win=5792 Len=0...
red	125	21.136783	203.0.113.0	192.0.2.1	TCP	54770->443 [SYN] Seq=0 Win=5792 Len=0...
red	126	21.459796	203.0.113.0	192.0.2.1	TCP	54770->443 [SYN] Seq=0 Win=5792 Len=0...
red	127	21.782809	203.0.113.0	192.0.2.1	TCP	54770->443 [SYN] Seq=0 Win=5792 Len=0...
red	128	22.105822	203.0.113.0	192.0.2.1	TCP	54770->443 [SYN] Seq=0 Win=5792 Len=0...
red	129	22.428835	203.0.113.0	192.0.2.1	TCP	54770->443 [SYN] Seq=0 Win=5792 Len=0...
red	130	22.751848	203.0.113.0	192.0.2.1	TCP	54770->443 [SYN] Seq=0 Win=5792 Len=0...
red	131	23.074861	203.0.113.0	192.0.2.1	TCP	54770->443 [SYN] Seq=0 Win=5792 Len=0...
red	132	23.397874	203.0.113.0	192.0.2.1	TCP	54770->443 [SYN] Seq=0 Win=5792 Len=0...
red	133	23.720887	203.0.113.0	192.0.2.1	TCP	54770->443 [SYN] Seq=0 Win=5792 Len=0...
red	134	24.0439	203.0.113.0	192.0.2.1	TCP	54770->443 [SYN] Seq=0 Win=5792 Len=0...
red	135	24.366913	203.0.113.0	192.0.2.1	TCP	54770->443 [SYN] Seq=0 Win=5792 Len=0...
red	136	24.689926	203.0.113.0	192.0.2.1	TCP	54770->443 [SYN] Seq=0 Win=5792 Len=0...
red	137	25.012939	203.0.113.0	192.0.2.1	TCP	54770->443 [SYN] Seq=0 Win=5792 Len=0...
red	138	25.335952	203.0.113.0	192.0.2.1	TCP	54770->443 [SYN] Seq=0 Win=5792 Len=0...
red	139	25.658965	203.0.113.0	192.0.2.1	TCP	54770->443 [SYN] Seq=0 Win=5792 Len=0...
red	140	25.981978	203.0.113.0	192.0.2.1	TCP	54770->443 [SYN] Seq=0 Win=5792 Len=0...
red	141	26.304991	203.0.113.0	192.0.2.1	TCP	54770->443 [SYN] Seq=0 Win=5792 Len=0...

red	196	44.070706	203.0.113.0	192.0.2.1	TCP	54770->443 [SYN] Seq=0 Win=5792 Len=0...
red	197	44.393719	203.0.113.0	192.0.2.1	TCP	54770->443 [SYN] Seq=0 Win=5792 Len=0...
red	198	44.716732	203.0.113.0	192.0.2.1	TCP	54770->443 [SYN] Seq=0 Win=5792 Len=0...
red	199	45.039745	203.0.113.0	192.0.2.1	TCP	54770->443 [SYN] Seq=0 Win=5792 Len=0...
red	200	45.362758	203.0.113.0	192.0.2.1	TCP	54770->443 [SYN] Seq=0 Win=5792 Len=0...
red	201	45.685771	203.0.113.0	192.0.2.1	TCP	54770->443 [SYN] Seq=0 Win=5792 Len=0...
red	202	46.008784	203.0.113.0	192.0.2.1	TCP	54770->443 [SYN] Seq=0 Win=5792 Len=0...
red	203	46.331797	203.0.113.0	192.0.2.1	TCP	54770->443 [SYN] Seq=0 Win=5792 Len=0...
red	204	46.65481	203.0.113.0	192.0.2.1	TCP	54770->443 [SYN] Seq=0 Win=5792 Len=0...
red	205	46.977823	203.0.113.0	192.0.2.1	TCP	54770->443 [SYN] Seq=0 Win=5792 Len=0...
red	206	47.300836	203.0.113.0	192.0.2.1	TCP	54770->443 [SYN] Seq=0 Win=5792 Len=0...
red	207	47.623849	203.0.113.0	192.0.2.1	TCP	54770->443 [SYN] Seq=0 Win=5792 Len=0...
red	208	47.946862	203.0.113.0	192.0.2.1	TCP	54770->443 [SYN] Seq=0 Win=5792 Len=0...
red	209	48.269875	203.0.113.0	192.0.2.1	TCP	54770->443 [SYN] Seq=0 Win=5792 Len=0...
red	210	48.592888	203.0.113.0	192.0.2.1	TCP	54770->443 [SYN] Seq=0 Win=5792 Len=0...
red	211	48.915901	203.0.113.0	192.0.2.1	TCP	54770->443 [SYN] Seq=0 Win=5792 Len=0...

red	212	49.238914	203.0.113.0	192.0.2.1	TCP	54770->443 [SYN] Seq=0 Win=5792 Len=0...
red	213	49.561927	203.0.113.0	192.0.2.1	TCP	54770->443 [SYN] Seq=0 Win=5792 Len=0...
red	214	49.88494	203.0.113.0	192.0.2.1	TCP	54770->443 [SYN] Seq=0 Win=5792 Len=0...
red	214	50.207953	203.0.113.0	192.0.2.1	TCP	54770->443 [SYN] Seq=0 Win=5792 Len=0...
red	214	50.530966	203.0.113.0	192.0.2.1	TCP	54770->443 [SYN] Seq=0 Win=5792 Len=0...
red	214	50.853979	203.0.113.0	192.0.2.1	TCP	54770->443 [SYN] Seq=0 Win=5792 Len=0...
red	214	51.176992	203.0.113.0	192.0.2.1	TCP	54770->443 [SYN] Seq=0 Win=5792 Len=0...
red	214	51.500005	203.0.113.0	192.0.2.1	TCP	54770->443 [SYN] Seq=0 Win=5792 Len=0...
red	214	51.823018	203.0.113.0	192.0.2.1	TCP	54770->443 [SYN] Seq=0 Win=5792 Len=0...

Diagnostic Findings

- **Attack Vector:** The attack vector is a volumetric TCP SYN Flood . The attacker repeatedly sends connection requests with zero data length (you'll see `Len=0` in the 'Info' column) to tie up server resources without completing the three-way handshake.
- **Impact:** High Severity. Complete denial of service for legitimate users, confirmed by TCP resets and HTTP 504 gateway timeouts.

Recommended Remediation Steps

1. **Deploy SYN Cookies:** Enable TCP SYN cookies on the target server to track connection states without consuming pool memory.
2. **Implement Firewall Rate Limiting:** Drop connections from external source IPs that exceed a reasonable `SYN` packet threshold.
3. **Configure a Reverse Proxy/WAF:** Position a Web Application Firewall or Content Delivery Network (CDN) in front of the server to absorb volumetric floods.